

- 1) show the operation of dual stack technique and explain its importance in transition from IPv4 to IPv6?

sol:

The dual stack technique is a method in which device runs both IPv4 & IPv6 at the same time. this allow the device to communicate with both IPv4 only and IPv6 network

Importance:

- Allows a smooth and gradual transition to IPv6
- Ensure backward Compatibility with IPv4
- Avoid Communication problem during migration.
- Helps with adopting IPv6 without supporting IPv4 service.

- 2) Using fixed length Subnetting divide the given network 196.220.84.32 into two parts. Give the first host Id, last host Id, subnetmask and number of host connected.

sol:

Given network 196.220.84.32 $\rightarrow$ Belongs to class c $n = 24$

so network = 196.220.84.32/24

Binary Format: 11000100.11011100.01010100.00100000

Subnetwork mask: 255.255.255.0

Divide into 2 parts $2^1 = 2$

$\rightarrow$ 11000100.11011100.01010100. 00100000 $\rightarrow$ 10010

case 1: bit = 0 $\rightarrow$ 196.220.84.32/25 } $n_1 = n_2 = 25$ Subnet = [255.255.255.128]
 case 2: bit = 1 $\rightarrow$ 196.220.84.160/25

1) First address (32-n): 196.220.84.0/25

2) Second address (32-n): 196.220.84.255/25

Subnet 1 $\rightarrow$ { 196.220.84.0/25
 range { 196.220.84.127/25

Subnet 2 $\rightarrow$ { 196.220.84.128/25
 range { 196.220.84.155/25

<u>Subnet 1</u>	<u>Subnet 2</u>
1) First Host id = $196.220.84.1/25$	1) $196.220.84.129/25$
2) Broadcast id = $196.220.84.127/25$	2) $196.220.84.255/25$
3) Last Host id = $196.220.84.126/25$	3) $196.220.84.254/25$
4) Subnet mask = $255.255.255.128$	4) $255.255.255.128$
5) no of Host = 126	5) 126.

3) Explain the function of following device a) Hub b) bridge c) Router d) Gateway!

1) Hub :->

A hub is a basic networking device used to connect multiple computers in a network. When data is received from one device, the hub broadcast is to all connected devices. It does not check the destination address and work at the physical layer.

2) Bridge :->

A bridge is used to connect two or more LAN segment. It filter traffic by checking MAC address and forward data only to the correct segment reducing network congestion.

3) Router :->

A router connect different network and forward data packets based on IP addresses. It select the best path for data transmission between network.

4) Gateway :->

A gateway is a device that connects networking using different protocols. It convert data from one protocol format to another so that communication is possible.

4) Give the network id and host id of the following IPv4

- a) 56.89.34.25 c) 212.115.65.24 e) 249.84.69.54.
b) 165.25.242.8 d) 227.54.68.26

Ip address	class	Network Id	Host id
56.89.34.25	A	56	89.34.25
165.25.242.8	B	165.25	242.8
212.115.65.24	C	212.115.65.24	
227.54.68.26	D	—	—
249.84.69.54	E	—	—

5) If the class B network on the internet has a Subnet mask of 255.255.245.0
What is the maximum number of host per subnet?

Sol: Given Subnet mask of 255.255.245.0

$$255.255.245.0 \rightarrow 11111111 \ 11111111 \ 11110000 \ 00000000 = 20 \text{ bit}$$

$$\text{No of address} = 2^{32-n} = 2^{32-20} = 2^{12} = 4096$$

$$\text{No of Host} = 4096 - 2 = 4094$$

6) An Organization is granted 32 blocks 211.14.130.0/24. The administrator want to create 32 subnet. a) Find subnet mask, b) Find the number of address in each subnet
Find the 1 & last in subnet 1, and subnet 32?

Sol: Given Ip address: 211.14.130.0/24

$$\text{Subnet: } 32 = 2^5 = 32 = \textcircled{5}$$

1) Subnet mask:

$$2^n = 32 \Rightarrow n = 5$$

$$\text{Ip address} = 211.14.130.0/24 \rightarrow \text{new length } 24 + 5 = 29$$

$$\text{Subnet mask: } 11111111 \ 11111111 \ 11111111 \ 11111000$$

$$\boxed{\text{Decimal dotted notation: } 255.255.255.248/29}$$

2) No of address: $2^{32-n} = 2^{32-29} = 2^3 = 8$

3) First and last address in subnet 1

• Network address - $211.14.180.0/29$ - 1st address - $211.14.180.1/29$

• Broadcast address - $211.14.180.4/29$ - Last address - $211.14.180.6/29$

4) First and last address in subnet 32

• Subnet number increment $(32-1) \times 8 = 248$

• Network address - $211.14.180.248/29$ - 1st address - $211.14.180.249/29$

• Broadcast address - $211.14.180.255/29$ - Last address - $211.14.180.254/29$

7) An ISP is granted a block of address starting with $190.100.0.0/16$ (65,536 address)

The ISP needs to distribute these address to three group customers as follows a)

The first group has 64 customers each need 256. b) The second group has 128 customer each need 128 address c) The third group need 128 customers each need 64 address

Design the subblock and find out how many address are still available after these allocation!

1) Group 1: 64 Customers each need 256

$$256 = 2^n = n = 8$$

The prefix length $(32-n) = 32-8 = 24$

1st Customer - $190.100.0.0/24$ $190.100.0.255/24$

2nd Customer - $190.100.1.0/24$ $190.100.1.255/24$

⋮
64th Customer - $190.100.63.0/24$ $190.100.63.255/24$

$$\text{Total} = 64 \times 256 = 16,384$$

2) Group 2: 128 Customer each need 128

$$128 = 2^n = n = 7$$

The prefix length $(32-n) = 32-7 = 25$

1st Customer - $190.100.64.0/25$ $190.100.64.127/25$

2nd Customer - $190.100.64.128/25$ $190.100.64.255/25$

128th customer 190.100.127.128/25

190.100.127.255/25

$$\boxed{\text{Total} = 128 \times 128 = 16,384}$$

Group 3: 128 customer 64 address

$$\text{address} = 64 = 2^6 = n = 6$$

$$\text{prefix length} = (32 - 6) = 26$$

$$1^{\text{st}} - 190.100.128.0/26 \quad 190.100.128.63/26$$

$$2^{\text{nd}} - 190.100.128.64/26 \quad 190.100.128.127/26$$

$$\vdots$$

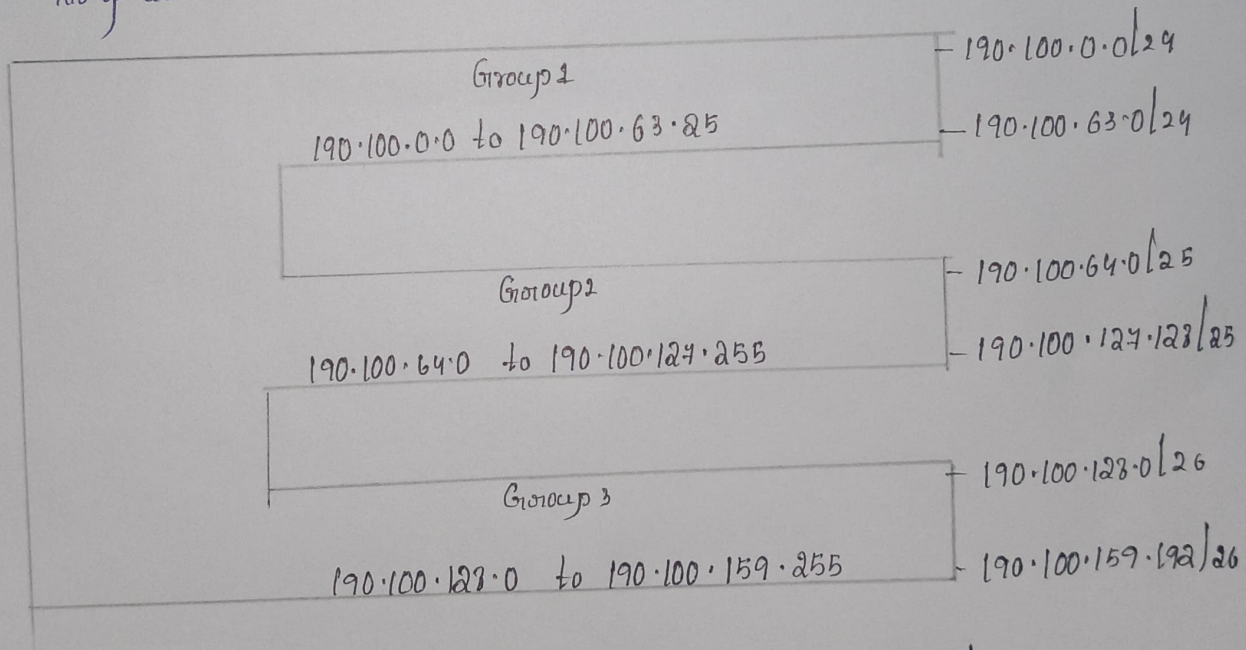
$$128^{\text{th}} - 190.100.159.192/26 \quad 190.100.159.255/26$$

$$\boxed{\text{Total} = 128 \times 64 = 8192}$$

No of granted address to Top: 65,536

No of allocated address by Top: 40,960

No of available address: 24,576



2) Suppose one of the address in a particular block is 192.63.110.114/24. Find the number of address that can be performed, the first address and the last address in the block.

sol: Given address 192.63.110.114/24

$$\text{No of address} = 2^{32-n} = 2^{32-24} = 2^8 = 256$$

$$\boxed{\text{address} = 256}$$

First address

→ 192.63.110.114

make right most bits to 0

$$\boxed{1^{st} = 192.63.110.0}$$

Last address

→ 192.63.110.114

make right most bits to 1

$$\boxed{2^{nd} = 192.63.110.255}$$